W

INTERVIEW PRESENTATION · 9 OCTOBER 2026

Building supplies, made simpler.

A role-aware commerce experience for customers, dealers, and plumbers, connected to Zoho Inventory.

Customer ordering | Verified dealer access | Plumber activities | Admin operations

One catalogue. Four role-aware experiences.

The product connects a trade-supply business to the people who buy, resell, and install its products.

Customers 01

Browse products, see retail price and availability, pay upfront, and track an order.

Dealers 02

Zoho-verified access, dealer pricing, exact stock, and admin-enabled payment terms.

Plumbers 03

Admin-approved sign-in, assigned activities, schedules, and service notifications.

Admins 04

Manage people, products, orders, payments, dealer credit, and reports.

Product principle: make role-specific data decisions on the server. Hiding a field in the app is not access control.

VALIDATED NOW OTP login · live Zoho contact checks · six live inventory items · server-side role filtering

Keep business rules in the backend.

WAYTOWEBS BACKEND

API + policy engine

OTP and role authorization
Zoho verification and catalog sync
Price, stock, credit, and order orchestration

PostgreSQL + Redis

Accounts, carts, orders, credit policy, OTP state and cache

Worker + durable queue

Scheduled sync, idempotent Zoho writes, retry and reconciliation

Mobile app

Customer · dealer · plumber

Admin web panel

Operations · credit · reporting

Zoho Inventory

Contacts · items · stock · sales documents

MSG91 + FCM

OTP delivery and order/activity notifications

Payment gateway

Mobile and web clients → HTTPS / JSON

Hosted payment and signed server webhook

External services; sync and write paths controlled by backend

04 · PROOF OF CONCEPT

Tested against the live Zoho account.

The risky integration questions have working end-to-end answers.

CATALOG

6 live Zoho items

Token refresh, item read, and five-minute cache implemented.

DEALER GATE

2 custom flags

Unique contact + Is Dealer + Verified checked before OTP.

ROLE BOUNDARY

2 product shapes

Exact stock only appears in the dealer response.

LIVE CHECK	OBSERVED RESULT	STATUS
Verified dealer OTP login	OTP issued; login returned role-bearing JWT	PASS
Dealer catalog	6 products each include dealer price and exact stock	PASS
Unverified dealer	Generic 403; no OTP issued; reason only in masked server log	PASS
Customer catalog	6 products include price + in_stock; stock field omitted	PASS

Demo OTP is console-only. No customer phone numbers, OTPs, JWTs, or credentials are shown on this slide.

Verify first. Reveal only what the role needs.

DEALER SIGN-IN

01 · Identify
Normalize mobile and validate role.

→

02 · Verify Zoho
Exactly one contact; Is Dealer + Verified true.

→

03 · Issue OTP
Only eligible dealer receives a one-time code.

→

04 · Authenticate
Consume OTP; sign JWT with role.

Failure behavior: generic dealer-facing denial. Detailed reason is server-side only; duplicate Zoho contacts are refused.

Customer / plumber

RETAIL

Retail price + `in_stock` availability flag.

No exact quantity or dealer-price field is returned.

Verified dealer

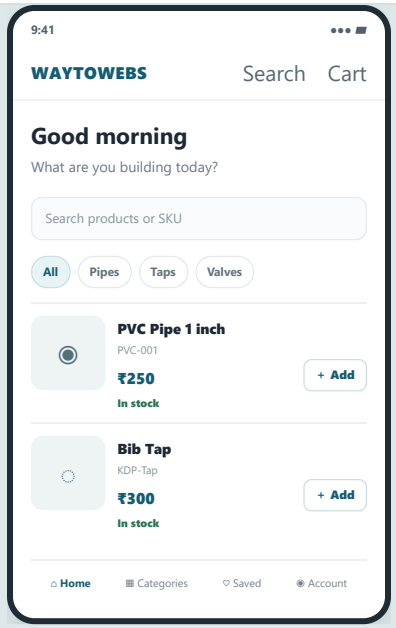
DEALER

Dealer price + exact `stock`, subject to authorized account.

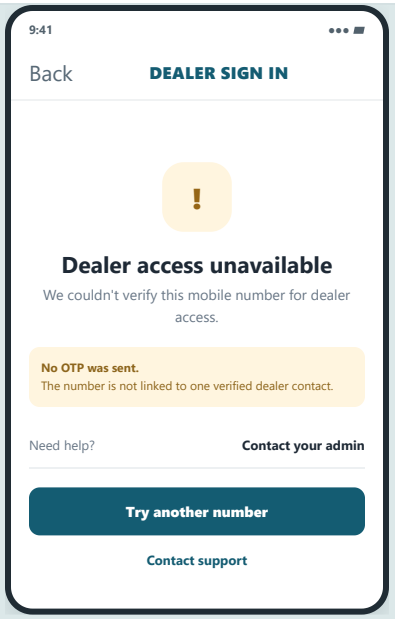
Demo currently uses illustrative 10% discount; production uses the agreed Zoho price list.

Designed around real role decisions.

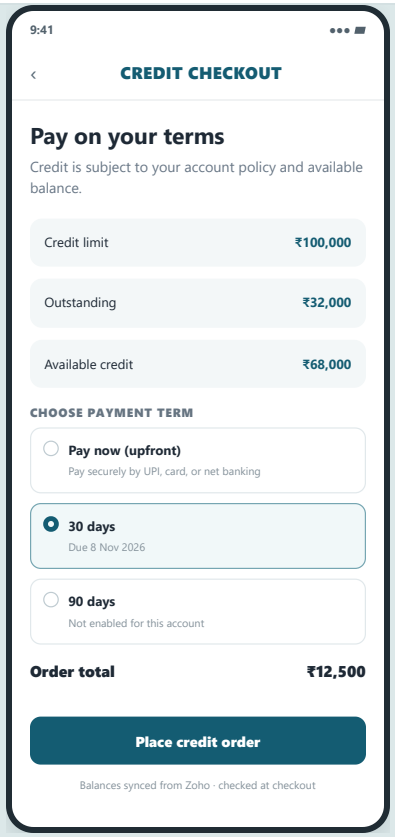
Customer · browse catalog



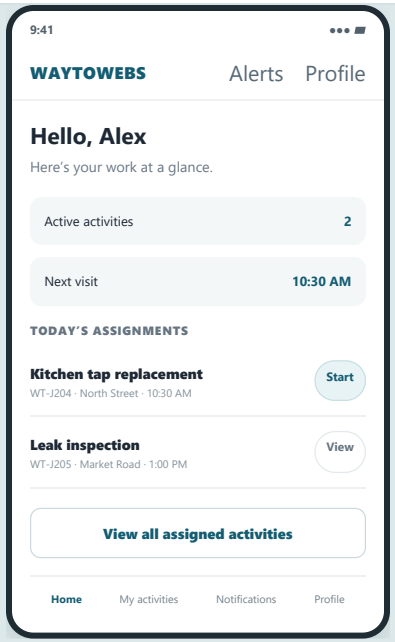
Dealer · Zoho rejection



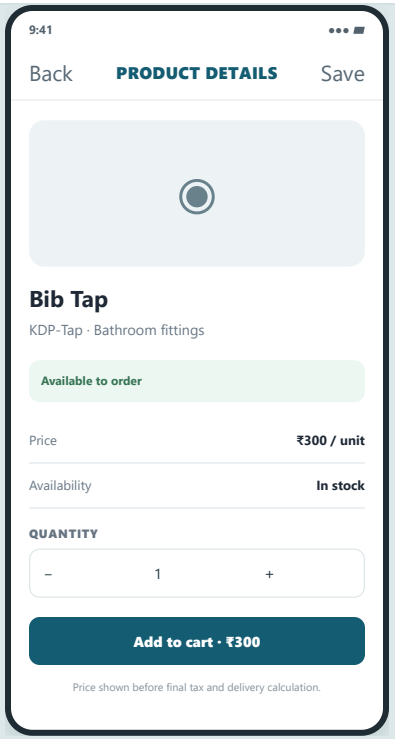
Dealer · credit terms



Plumber · dashboard



Customer · product detail



Selected screens from the full role-based gallery · fictional names and sample values.

Different roles. One consistent source of truth.

CUSTOMER VS DEALER

FEATURE	CUSTOMER	DEALER
Login	OTP; account creation proposed	OTP after unique verified Zoho match
Price	Retail	Authorized dealer price list
Stock	Availability flag only	Exact quantity
Payment	Upfront	Upfront; admin-enabled 30/90-day terms
Credit	None	Admin-set limit; balance checked at checkout
Zoho order	After verified payment	After payment or approved credit

ADMIN WEB PANEL · EIGHT MODULES

- **01** Customers
- **02** Dealers
- **03** Plumbers
- **04** Products
- **05** Orders
- **06** Payments
- **07** Credit
- **08** Reports

Admin approves plumbers and configures dealer credit policy. Zoho remains authoritative for live inventory and receivables.

Zoho owns operational records; the app owns policy.

DATA	DIRECTION	PROPOSED CADENCE	ZOHO API SURFACE TO VALIDATE
Items, retail prices, stock	Zoho → app DB/cache	Every 5–10 min + supported webhook	`GET /items`
Dealer price list	Zoho → app DB/cache	With item sync	Price list API
Dealer contacts + verification flags	Zoho → backend	Live at dealer login; optional nightly sync	`GET /contacts`, `GET /contacts/{id}`
Outstanding + overdue invoices	Zoho → app DB	Hourly and before credit checkout	`GET /contacts/{id}`, `GET /invoices`
New customer record	Backend → Zoho	First order, if required by workflow	`POST /contacts`
Sales order / credit invoice	Backend → Zoho	After payment or credit approval	`POST /salesorders` / invoice API
Payment record	Backend → Zoho	After verified gateway webhook	`POST /customerpayments`

Resilience: durable queue + idempotent order reference for Zoho writes. If Zoho is unavailable, retain and retry sync; deny credit checkout if a fresh balance cannot be verified.

Cadences and endpoint names are planning assumptions to confirm against the selected Zoho product, data center, scopes, and API version.

Small working core. Clear production roadmap.

Implemented and smoke-tested

EXPRESS POC

POST `/auth/send-otp`
Public · customer, dealer, plumber eligibility

POST `/auth/verify-otp`
Public · consumes OTP; returns JWT

GET `/products`
JWT · role-filtered catalog

GET `/me`
JWT · current identity/claims

GET `/`
Public · health check

Proposed next

NOT IMPLEMENTED

GET `/categories`, `/products/:id`
Authenticated · discovery and detail

GET/PUT `/cart`
Authenticated · server-priced cart

POST `/orders`
Customer/dealer · client sends item IDs, quantities, term only; server resolves prices/stock

POST `/payments/webhook`
Gateway signature · server-to-server confirmation

GET `/dealer/credit`
Dealer · enabled terms and available balance

GET `/plumber/activities`
Approved plumber · assigned work

/admin `/admin/*`
Admin-only account and operations APIs

Order integrity: ignore client-supplied price or stock. Reprice, recheck availability and credit server-side; apply ownership checks, idempotency keys, and signed payment webhooks.

Detailed request/response notes: api-endpoints.md

Payment verified by webhook. Credit checked fail-closed.

Customer · upfront

PROPOSED

1 · Checkout
Server rechecks item price, availability, GST and delivery.

↓

2 · Pay
Gateway handles payment; signed webhook confirms it to backend.

↓

3 · Fulfil
Idempotent Zoho order job; retry if Zoho is temporarily down.

Dealer · upfront / terms

PROPOSED

1 · Select term
Pay now, or admin-enabled 30/90-day credit.

↓

2 · Check policy
Admin-set limit + term; Zoho-synced outstanding + overdue.

↓

3 · Create document
`outstanding + order ≤ limit`; due date included in Zoho write.

Policy ownership: admin sets credit limits and allowed terms in the app database; Zoho is authoritative for outstanding/overdue balances. If balance data is stale or unavailable, reject credit checkout. Queue and retry failed Zoho document creation.

Credit rejection cases: term not enabled · limit exceeded · overdue invoices (if policy blocks) · Zoho balance unavailable/stale. No credit is approved on uncertainty.

Sequence the work around integration risk.

WEEK 1
Requirements & design
Flows, roles, assumptions

WEEKS 2–3
Backend & auth
OTP, roles, data model

WEEKS 3–4
Zoho sync
Contacts, catalog, stock

WEEKS 4–6
Customer commerce
Catalog, cart, payment

WEEKS 6–7
Dealer credit
Pricing, policy, receivables

WEEK 8
Plumber module
Approval and activities

WEEKS 8–9
Admin & reports
Ops and account controls

WEEKS 10–11
QA & release
Security, UAT, launch

Indicative 10–12 weeks with planned overlap. Validate team size and Zoho dependencies before commitment.

RISK

Zoho limits / downtime
Cache reads; backoff queue; reconciliation.

RISK

Duplicate contacts
Reject ambiguous match; admin data repair.

RISK

OTP abuse
Rate limits, attempt caps, expiry; move to Redis.

RISK

Payment/order mismatch
Signed webhook, idempotency, reconciliation.

Decisions to confirm before committing scope.

Zoho product and region

The brief says “Zoho Workbook,” while the current connector uses Zoho Inventory. Which product, data center, and organization are expected?

Dealer pricing

One shared dealer price list, or different lists/discounts by dealer?

Credit ownership

Can admin-set limits/30- and 90-day terms live in our DB while balances come from Zoho?

Credit rules

Are overdue invoices a hard block? How fresh must balance sync be at checkout?

Plumber verification

Registration details, ID documents, approval authority, and rejection process?

Plumber job scope

Do plumbers see assigned jobs, buy supplies, or both? Who assigns and updates status?

Payments and messaging

Preferred gateway (e.g. Razorpay), SMS provider (e.g. MSG91), and refund flow?

Delivery and tax

Who fulfils orders and owns tracking? How are GST and delivery charges calculated?

> **Working assumptions:** customer pays upfront; dealer may pay upfront or use admin-enabled credit; orders sync to Zoho only after verified payment or approved credit. Confirm these with the interviewer.

PROTOTYPE VALIDATED

PRODUCTION PLAN IS A PROPOSAL